Arthur Berberyan

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Education

University of California, San Diego (UCSD), La Jolla, CA, USA
Ph.D. Student in Astronomy & Astrophysics — September 2025 – Present
Advisor: Dr. Christopher A. Theissen

California State University, Northridge (CSUN), Northridge, CA, USA
M.S. in Physics (with Distinction) — August 2023 – May 2025
Thesis: *Mode Coupling of Magnetic Bright Points in the Solar Atmosphere*
Sigma Pi Sigma Honor Society

B.S. in Physics (cum laude) — August 2021 – May 2023
Specialization: Astrophysics
Advisor: Dr. Damian J. Christian

Publications

Keys, P. H., Campbell, R. J., Magill, D. K. J., et al. 2026, *Small-scale bright point characteristics at high-resolution with the Daniel K. Inouye Solar Telescope*, ApJ, 999, 201.

Berberyan, A., Keys, P. H., Jess, D. B., & Christian, D. J. 2024, *A search for mode coupling in magnetic bright points*, A&A, 690, A363.

Clark, J. T., Addison, B. C., Okumura, J., et al. 2023, *Spinning up a Daze: TESS Uncovers a Hot Jupiter Orbiting the Rapid Rotator TOI-778*, AJ, 165, 207.

Christiansen, J. L., Bhure, S., Zink, J. K., et al. 2022, *Scaling K2. V. Statistical Validation of 60 New Exoplanets from K2 Campaigns 2–18*, AJ, 163, 244.

Presentations & Conferences

The search for star-planet interactions in exoplanet systems with highly elliptical orbits. 245th annual American Astronomical Society (AAS), National Harbor, Maryland. [iPoster] (January 2025)

Searching for mode coupling in magnetic bright points. CSUN Department of Physics & Astronomy Colloquium, Northridge, CA. [slides] (September 2024)

The Search for Mode Coupling in the Lower Solar Atmosphere. 22nd Cambridge Workshop on Cool Stars, Stellar Systems, and the Sun, University of California, San Diego, CA. [DOI] (June 2024)

The Recent Solar Eclipse and Understanding Mode Coupling in Our Sun. Association of Retired Faculty (ARF), Northridge, CA. [slides] (May 2024)

A Search for Mode Coupling in Magnetic Bright Points. Department of Science & Mathematics, CSUNposium, Northridge, CA. (May 2023)

Elastic Wave Propagation in Compressed Phononic Crystals, REU. Princeton, NJ. [poster], [report] (August 2022)

Research Experience

CSUN Solar Physics Group with Dr. Damian J. Christian — August 2021 – May 2025
I investigated the coronal heating issue by studying mode coupling in magnetic bright points within the different layers of the Sun to better understand how energy is transported to heat the chromosphere and the overall energy budget.

Princeton University Summer REU with Dr. Andrej Košmrlj — June 2022 – August 2022
At the Center for Complex Materials (PCCM), I participated in work involving Acoustic Metamaterials. I looked into elastic wave propagation in compressed phononic crystals, and how they behaved in deforming materials.

IPAC (Caltech) Research Assistant with Dr. David R. Ciardi — August 2019 – August 2020
Using data from NASA's TESS Satellite, Palomar, and Keck Observatories, I categorized 150+ TESS Objects of Interest (TOI) stars with exoplanets alongside high-resolution image processing with near-infrared adaptive optics.

Palomar Observatory Research Assistant June 2019
Collaborated with Caltech astronomers imaging star systems with candidate exoplanets using the 200-inch Hale telescope.

Teaching

UC San Diego, Department of Astronomy & Astrophysics:
ASTR 15: Astronomy in Science Fiction (Discussion) March 2026 – June 2026
ASTR 124: Exoplanets (Discussion) September 2025 – December 2025

CSUN, Department of Physics & Astronomy:
ASTR 154L Observational Astronomy Lab August 2023 – May 2025
PHYS 220AL Mechanics Lab January 2024 – May 2024
PHYS 100B General Physics II August 2023 – December 2023

Awards & Honors

2025 – 2026 UCSD Astronomy & Astrophysics Achievement Award – $2,000
2025 Sigma Pi Sigma Honor Society Member, American Institute of Physics
2024 Department of Physics & Astronomy Travel Award – $1,500
2024 Department of Physics & Astronomy Summer Research Award – $4,000
2024 The Debra Costa Graduate Student Scholarship – $1,500
2024 Association of Retired Faculty Award – $2,500
2023 CSUNposium awardee in the Department of Science & Mathematics – $250
2023 Betty and Martin Altshiller Memorial for Outstanding Undergraduates – $1,000
2023 Paul and Amy Lee Undergraduate Scholarship for Excellence – $1,500
2022 – 2023 CSUN Undergrad Department of Science & Mathematics Dean's List
2021 MSP SAGE Society Scholarship – $1,200
2019 California Space Grant Consortium – $1,000
2018 NASA Community College Aerospace Scholar (NCAS)

Outreach & Service

UCSD A&A Peer Mentorship, *Mentor (2 Undergraduates)* October 2026 – Present
CSUN Star Party, *Celestron NexStar 8SE (8") Telescope Operator* March 2025
CSUN Society of Physics Students, *Member* August 2021 – May 2025
CSUN Science Day Fair, *Experimental Demonstrator* May 2023, 2024
CSUN Underrepresented Students in STEM Program, *Mentor* June 2023
Princeton Partnership with Trenton High School Students, *Mentor* June – July 2022

References

Dr. Christopher A. Theissen
Assistant Professor of Astronomy & Astrophysics at UCSD, **Email:** ctheissen@ucsd.edu.

Dr. Damian J. Christian
Professor of Physics & Astronomy at CSUN, **Email:** damian.christian@csun.edu.

Dr. Andrej Košmrlj
Associate Professor of MAE at Princeton, **Email:** andrej@princeton.edu.

Dr. David R. Ciardi
Acting Executive Director of NExScI at IPAC (Caltech), **Email:** ciardi@ipac.caltech.edu.